Name_____ Date_____ Period_____

LESSON 7-6 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Distance Between Points on a Grid

Maryland Standard 6.NS.C.8 · Learning target – I can find the distance between two points that share a coordinate.

REMEMBER

- **Same x-coordinate**: the points are on a vertical line. Use the y-coordinates.
- **Same y-coordinate**: the points are on a horizontal line. Use the x-coordinates.
- Opposite signs: **add** the absolute values. Same signs: **subtract** them.

WORKED EXAMPLE

Find the distance from (3, −4) to (3, 5).

Step 1 Same x, so use the y-coordinates −4 and 5.

Step 2 Opposite signs, so add: 4 + 5 = 9.

Answer **9 units**

WORD BANK

distance – how far apart; never negative

x-coordinate – the first number; left or right

y-coordinate – the second number; up or down

absolute value – how far a number is from 0

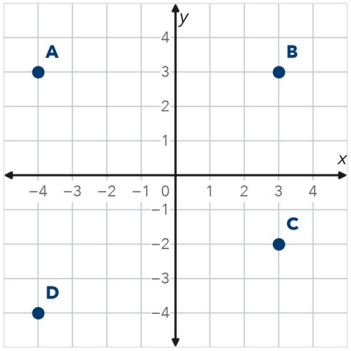
PRACTICE

show your work · label units

1. Use the grid. Find each distance.

Step 1 Find the coordinate that stays the same.

Step 2 Look at the other coordinate. Different signs? **Add**. Same sign? **Subtract**.



a. A to B _____ units

b. B to C _____ units

c. A to D _____ units

2. Find the distance between each pair of points.

a. (5, 6) and (5, −2) ____

b. (−7, 1) and (−2, 1) ____

c. (−3, −4) and (6, −4) ____

d. (0, −8) and (0, 3) ____

e. (−9, −2) and (−9, −7) ____

f. (2.5, 3) and (−4, 3) ____

g. (−1, −6) and (−1, 4) ____

h. (−8, 5) and (−3.5, 5) ____

3. **City Blocks.** Kim walks **straight** from (−6, −3) to (−6, 5). Each unit is 1 block. How far does she walk?

   Step 1 The x-coordinates match, so use the y-coordinates −3 and 5.

   Step 2 The signs are different, so **add** the absolute values.

   Answer_____ blocks

4. **Missing Point.** Point P is at (2, −3). Point Q is 6 units to the **left** of P. Name point Q.

   Step 1 Moving left changes only the x-coordinate.

   Step 2 Subtract 6 from the x-coordinate. Keep the y-coordinate.

   Answer_____

5. **Find the Error.** Eli says the distance from (1, −6) to (1, −2) is 8 because 6 + 2 = 8. Explain his mistake and find the correct distance.

   Frame *Eli is wrong because both y-coordinates are ___, so he should ___. The distance is ___.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME With someone at home, count the blocks between two places on one street.

Name_____ Date_____ Period_____

LESSON 7-6 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Comparing Distances on the Plane

Maryland Standard 6.NS.C.8 · Learning target – I can decide whether to add or subtract to find a distance, then compare two distances.

REMEMBER

- Find **each** distance first. Then compare.
- Different signs: **add**. Same sign: **subtract**.
- A distance is never negative.

WORKED EXAMPLE

Which is longer: (−2, 4) to (5, 4), or (1, −3) to (1, 5)?

Step 1 First: same y, use x: 2 + 5 = 7 units.

Step 2 Second: same x, use y: 3 + 5 = 8 units. 8 > 7.

Answer **the second one**

WORD BANK

compare – decide which is greater

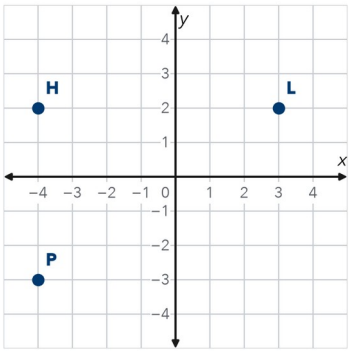
farther – a longer distance

closer – a shorter distance

PRACTICE

show your work · label units

1. Neighborhood. Each unit is 1 block. Which is closer to Home: the Library or the Park?



Step 1 Find H to L. Then find H to P.

Step 2 Compare the two numbers. Subtract to find how many more.

a. H to L _____ blocks

b. H to P _____ blocks

c. Which is closer? By how many blocks? _____

2. Find each distance.

a. (−8, 1) to (−3, 1) ____

b. (4, −6) to (4, 7) ____

c. (−6, −7) to (−6, −1) ____

d. (−2.5, 3) to (4.5, 3) ____

3. Runners. Ana runs from (−5, −1) to (6, −1). Jay runs from (2, 4) to (2, −6). Who runs **farther**? By how many units?

Step 1 Find Ana's distance.

Step 2 Find Jay's distance.

Step 3 Subtract the smaller distance from the larger one.

Answer_____

4. **Reasoning.** For (−3, 7) to (−3, −2), do you add or subtract? Explain, then find the distance.

Frame *I ___ because the y-coordinates 7 and −2 have ___ signs. The distance is ___ units.*

TEST PRACTICE

choose the best answer

5. Segment JK runs from (−4, 2) to (6, 2). Segment MN runs from (1, −3) to (1, 5). Which is longer?

Step 1 Find the length of JK.

Step 2 Find the length of MN. Then compare.

A JK, by 2 units **B** MN, by 2 units **C** neither **D** JK, by 8 units

6. Point A is at (−2, −5). Point B is at (−2, 3). What is the distance from A to B?

A 2 units **B** 3 units **C** 8 units **D** −8 units

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME With someone at home, pick two places on one street. Which is closer to your home? By how many blocks?